CRYSTAL GROWTH & DESIGN

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Bulk Crystal Growth, First-Principles Calculation, and Optical Properties of Pure and Er^{3+}-Doped $SrLaGa_3O_7$ Single Crystals

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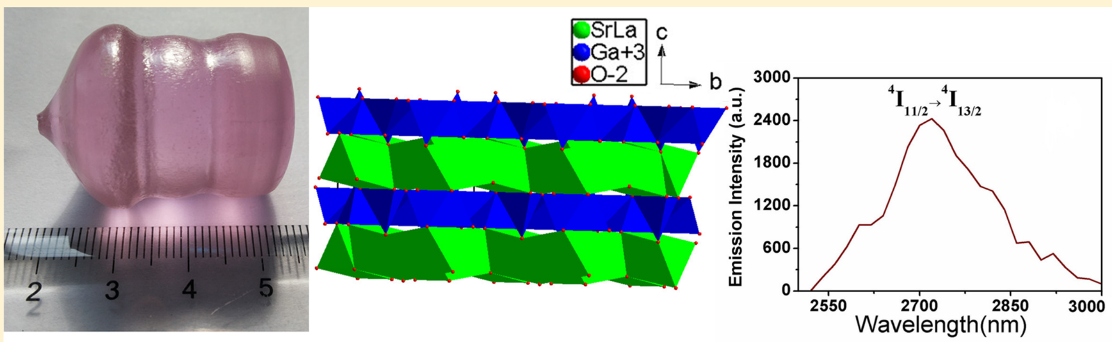
S Supporting Information



ABSTRACT: High-quality bulk single crystals of pure and Er^{3+}-doped $SrLaGa_3O_7$ (SLGO) were grown successfully by the Czochralski method. The crystal structure of SLGO was demonstrated. The theoretical calculations based on density functional theory methods were carried out on the SLGO crystal. Its band structure and density of state were presented. The absorption, up-conversion, near-infrared and mid-infrared fluorescence spectra, as well as the fluorescence decay curves of Er: $^4I_{13/2}$ and $^4I_{11/2}$ levels of the Er:SLGO crystal were measured at room temperature. The spectroscopic properties including the absorption and emission cross sections as well as the fluorescence lifetimes were revealed. It is concluded that the Er:SLGO crystal is a promising candidate material for achievement of a mid-infrared laser.

INTRODUCTION

In recent years, mid-infrared solid-state lasers at the wavelength region of 2.7−3.0 μm have attracted great attention for numerous applications in medical surgery, dentistry, and remote sensors, etc., mainly due to the vibration absorption band of water in this spectral region.[1−3] They also can be used in environmental protection, atmosphere detection, and military fields, etc. Besides, 2.7−3.0 μm lasers are suitable pump sources for mid-infrared or far-infrared optical parametric oscillation (OPO) or optical parametric generation.[4,5] Er^{3+} can serve as an active ion for emitting 2.7−3.0 μm by $^4I_{11/2} \rightarrow {}^4I_{13/2}$ transition, and more recently, a lot of work has been focused on Er^{3+} activated laser crystals.[1−3,6−11]

$SrLaGa_3O_7$ (SLGO) belongs to the large group of crystals with the chemical formula ABC_3O_7, where A = Ca, Sr, Ba; B = La, Gd; and C = Ga, Al crystallizing in the tetragonal melilite structure with space group $P\overline{4}2_1m$. Its lattice parameters are a = 8.06 Å, c = 5.34 Å. The density is 5.251 g/cm^3. SLGO melts congruently at 1588 °C and may be obtained in a single crystal formed by the Czochralski method.[12]

The earliest crystal growth of SLGO might have begun in 1992 by Polish scientist Ryba-Romanowski et al.[13] It is an excellent laser gain media, and up until now, many kinds of rare earth ions were doped into this crystal and studied systematically, such as Nd^{3+},[14−16] Pr^{3+},[17] Dy^{3+},[18] Ho^{3+}, etc.;[19−22] a Co^{2+}-doped SLGO crystal was studied with focus on the Jahn−Teller effect.[23,24] The studies on the nonlinear refractive indices and the radiation defects of SLGO crystal were carried out.[25−27] The crystal-field analysis for Nd^{3+} ions in SLGO crystal was demonstrated.[28] The whole-view research on the analyses of broad band, near-infrared emission in $ABCO_4$ and ABC_3O_7 crystals was presented by the earliest researcher of this crystal (Witold Ryba-Romanowski) in 1997.[29] High-optical-quality 1 at % Nd^{3+}-doped SLGO crystal with a length up to 50 mm was obtained by Czochralski method, and the piezoelectric, elastic, and dielectric constants of it were determined. It was concluded that SLGO is an excellent high temperature piezoelectric crystal.[12,30] Besides, Kaminskii et. al reported the first observation of stimulated Raman scattering (SRS) effects in SLGO crystal recently.[31]

In the SLGO crystal, rare earth ions can substitute for both the Sr^{2+} and the La^{3+} ions that are randomly distributed in a 1:1 ratio, producing a disordered structure and a large inhomogeneously broadening of both absorption and emission spectra. The broadening of the absorption band is very favorable for diode pumping, and it allows for higher tolerance in pumping wavelength, relaxing the requirements of diode wavelength

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control. On the other hand, the broadening of the emission band is beneficial for tunable or ultrafast laser output. In 2013, the transform-limited pulses as short as 378 fs around 1061 nm and pulses as short as 534 fs at 1071 nm were achieved in a Nd:SLGO crystal, pumped by a Ti:sapphire laser tuned to 808 nm and passively mode-locked with a SESAM; it was reported as the shortest pulses achieved with SESAM mode-locking in disordered Nd^{3+}-doped crystals.[32] However, owing to the limit of crystal quality and laser technology, the mid-infrared waveband laser output of the Er^{3+} activated SLGO crystal has not been realized yet, and this crystal could also act as a 2.7−3.0 μm ultrafast laser gain media.

In this paper, the growth of SLGO and Er:SLGO bulk crystals was reported, and the crystal structure of SLGO was discussed. Also, the first-principles calculations on SLGO crystal are carried out for the first time. The optical properties of the Er:SLGO crystal were presented too.

■ EXPERIMENTAL SECTION

Synthesis. Polycrystalline samples of SLGO and 20 at % Er^{3+}-doped SLGO were synthesized by using the traditional solid-state reaction techniques. The used raw materials were $SrCO_3$ (Alfa Aesar Chemical Reagent Co., Ltd., 99%), La_2O_3 and Er_2O_3 (Changchun Highpurity Chemical Reagent Co., Ltd., 99.99%), and Ga_2O_3 (Shanghai Mingqin Chemical Reagent Co., Ltd., 99.99%) powders. The compounds were dried and then weighed out according to the compositional formula, with an excess of Ga_2O_3 (1.0 wt %) added to the starting components for compensation for the evaporation of Ga_2O_3 during the growth process. The mixtures were ground, mixed, and pressed into tablets. The tablets were put into a platinum (Pt) crucible in a muffle furnace, then heated to 1000 °C at the rate of 150 °C/h and maintained for 20 h to completely decompose the $SrCO_3$, and then heated to 1150 °C at the rate of 100 °C/h and maintained for 10 h. The muffle furnace was shut down and cooled down to room temperature naturally. The sintered tablets were taken out and milled carefully, and then pressed into tablets again. The tablets were sintered at 1250 °C for 10 h to synthesize polycrystalline compounds. Generally, the above sintering processes were continuously carried out, until the pure-phase SLGO and Er:SLGO polycrystalline materials can be obtained, which were confirmed by X-ray diffraction (XRD, Miniflex600) using Cu Kα (λ = 1.54056 Å) with a graphite monochromator at room temperature.

Single Crystal Growth. Crystals were grown in an iridium (Ir) crucible by using the conventional radio frequency (RF)-heating Czochralski method in an atmosphere of N_2. The SLGO and Er:SLGO compounds were melted and kept for 2 h at a temperature 50 °C above the melting point, respectively, to ensure homogeneity of the melt. The *a*-cut SLGO crystal seed was used in the crystal growth experiment. The neck pulling rate was set at 0.6−2.0 mm/h, and then in the diameter stage, the pulling rate was set as 1.0−2.5 mm/h with a rotation rate of 12−20 rpm for the crystal growth. After the growth was completed, the crystal was pulled out of the melt and cooled slowly to room temperature at a rate of 8−25 °C/h. Finally, a transparent colorless crystal (SLGO) and a pink crystal (Er:SLGO) were obtained as shown in Figure 1; the bottom part of the Er:SLGO crystal is shown in Figure S1.

Powder X-ray Diffraction. Powder X-ray diffraction of the grown crystals was performed on a Miniflex600 X-ray diffractometer equipped with a diffracted beam monochromator set for Cu Kα radiation (λ = 1.54056 Å) in the 2θ range of 10−70° with a step size of 0.02° and a step time of 0.4 s at room temperature.

Ions' Concentration Analysis. The concentrations of Er^{3+}, La^{3+}, Sr^{2+}, and Ga^{3+} in the grown Er:SLGO crystal were measured by inductively coupled plasma atomic emission spectrometry (ICP-AES, Ultima2, Jobin-Yvon).

Crystal Structure Analysis. Single crystal X-ray diffraction data collections for SLGO were output from the ICSD database (No. 402556).

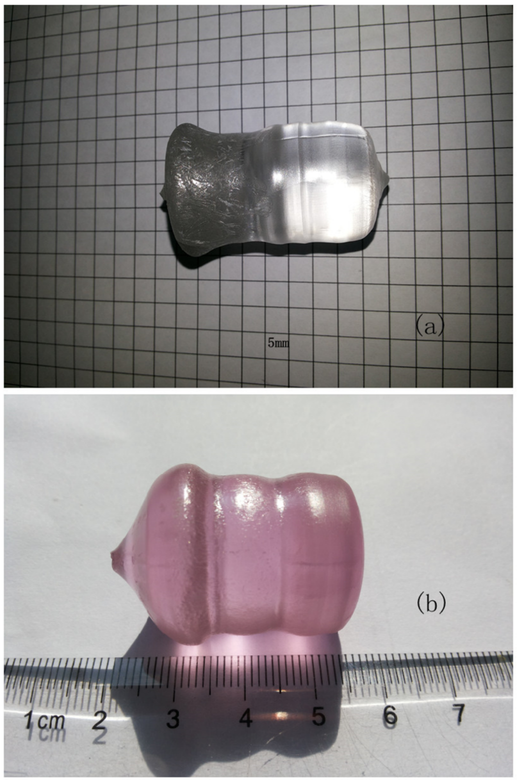

Figure 1. Photos of grown crystals: (a) $SrLaGa_3O_7$; (b) Er:$SrLaGa_3O_7$.

Optical Properties. One sample along the [100] direction (*a* axis) was cut from the as-grown Er:SLGO crystal with dimensions of 5.0 × 5.0 × 1.0 (*a* × *c* × *a*) mm^3. The two 5.0 × 5.0 mm^2 faces were polished for using in spectral experiments, as shown in Figure S2. The absorption spectrum of it was measured by a PerkinElmer UV−vis−NIR spectrometer (Lambda-950). The near-infrared (NIR) fluorescence spectrum and the decay curve were recorded in an Edinburgh Instruments FLS980 spectrophotometer by using a 980 LD pump source. The up-conversion (UC) fluorescence spectrum was measured by using an Edinburgh Instruments FLS920 spectrophotometer with a 980 LD pump source too. The mid-infrared (MIR) fluorescence spectrum and the decay curve were recorded by an Edinburgh Instruments FSP920c spectrophotometer, under 980 nm pumping with a 5 ns pulse of an optical parametric oscillator. All the measurements were carried out at room temperature.

Computational Descriptions. Single crystal structural data of SLGO were used for the theoretical calculations. The electronic structures including band structure and density of states (DOS) calculations were performed by using a plane-wave basis set and pseudopotentials within density functional theory (DFT) implemented in the total-energy code CASTEP. The exchange and correlation effects were treated by Perdew−Burke−Ernzerhof (PBE) in the generalized gradient approximation (GGA). The interactions between the ionic cores and the valence electrons were described by the norm-conserving pseudopotential. The following valence-electron configurations were considered in the computation: Sr-$4s^24p^65s^2$, La-$5d^16s^2$, Ga-$3d^{10}4s^24p^1$, and O-$2s^22p^4$. The number of plane waves included in the basis sets was determined by the cutoff energy of 600 eV. In addition, the numerical integration of the Brillouin zone was performed by using a Monkhorst−Pack *k*-point sampling of 2 × 2 × 2. The other parameters and convergence criteria were the default values of the CASTEP code.

■ RESULTS AND DISCUSSION

Crystal Growth and XRD. Photos of the obtained crystals are shown in Figure 1. The pure SLGO crystal is colorless and transparent; the upper part of it is a little dark owing to the

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evaporation of Ga_2O_3, which is then attached on the surface of the grown crystal. The grown Er:SLGO crystal is pink with good optical quality. Figure 2 shows the XRD pattern of the Er:SLGO crystal powder, and it is well consistent with the standard JCPDF file [No. 45-0637] for SLGO crystal, which means that the grown crystal is exactly pure phase. The doping concentrations of Er^{3+}, La^{3+}, Sr^{2+}, and Ga^{3+} in the Er:SLGO crystal were measured to be 2.29×10^{20}, 5.52×10^{21}, 5.50×10^{21}, and 1.58×10^{22} ions·cm^{-3}, respectively.

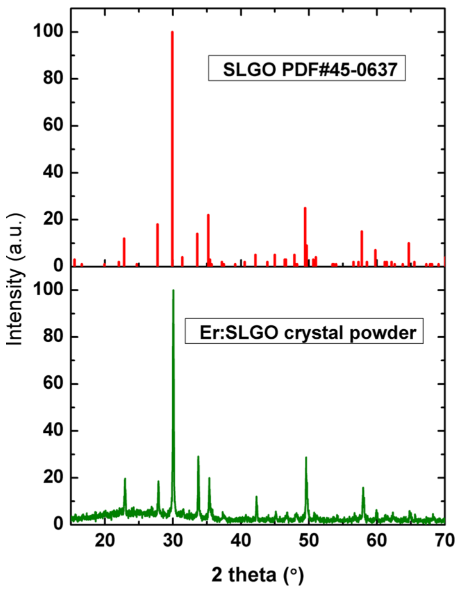

Figure 2. X-ray diffraction patterns of $SrLaGa_3O_7$ and Er: $SrLaGa_3O_7$ crystals.

Crystal Structure. Good optical properties are often related to the crystal structure due to these important aspects, e.g., crystallization dependent on anisotropy of crystallographic frame,[33,34] luminescence dependent on crystallization and electronegativity of dopant ions in the crystal lattice.[35−37] In order to further understand the crystal structure and substitution of Er^{3+} ion in the SLGO host, we present the crystal structure of SLGO in Figure 3. As seen in Figure 3, the crystal structure is constructed from large polyhedral layers alternating along the *c* axis, forming 5-fold coordination rings from GaO_4^{5-} tetrahedra linked at each corner. These large sites have a much distorted square antiprism configuration, known as Thomson cubes.[38] In SLGO, these tubes are filled statistically by Sr^{2+} and La^{3+} in the ratio of 1:1. All of the tetrahedra are filled by Ga^{3+} ions, which possess two different types of tetrahedra in the lattice. One has S_4 symmetry; the other has mirror symmetry. The latter tetrahedra are distorted and connected to each other at one corner, forming pairs of pyramids and having one vertex along the optical axis. These tetrahedra are liable to form a dominant C_{3v} distortion. A weaker perturbation reduces the local symmetry to C_s.[38] The

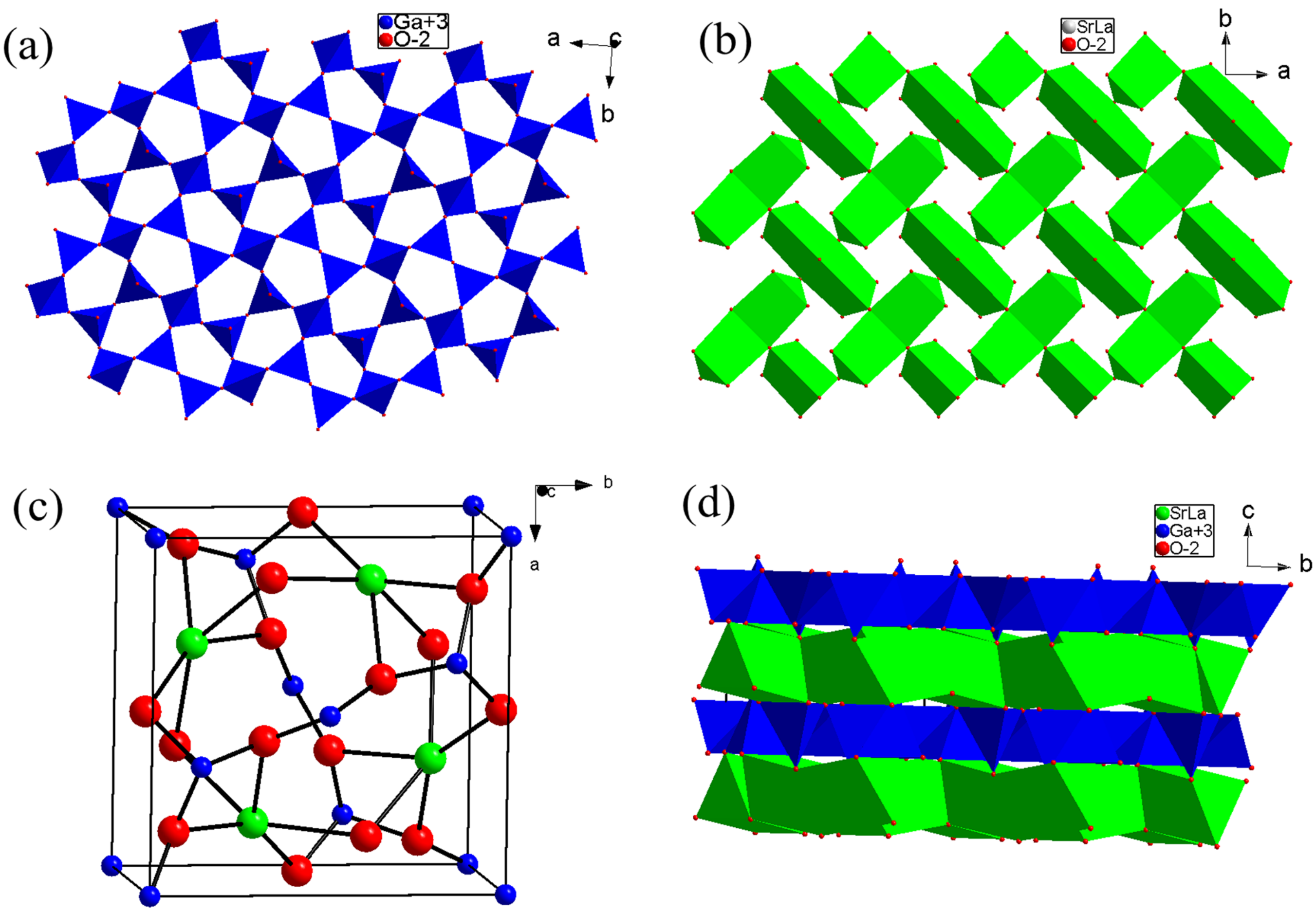

Figure 3. Crystal structure of $SrLaGa_3O_7$: coordination conditions of Ga-O in the *a*−*b* plane (a), coordination conditions of SrLa-O in the *a*−*b* plane (b), projection of SLGO unit cell in the *a*−*b* plane (c), and the layer structure along the *a* axis (d).

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distribution of La^{3+} and Sr^{2+} ions is random, and the large size disparity of La^{3+} and Sr^{2+} leads to significant lattice distortion.

Once Er^{3+} was doped into this crystal, three kinds of ions including Er^{3+}, La^{3+}, and Sr^{2+} are situated in one crystal lattice randomly and thus makes the crystal more disordered. In the unit cell ($Z = 2$) of SLGO, Sr and La atoms occupy only the 4e positions. Half of the 4e sites are expected to be filled by Sr atoms, and the other half by Er/La atoms. Sr/La sites corresponding to polyhedra with 8-coordinations are likely to be substituted by Er ions. The lattice disorder is highlighted in the Er:SLGO crystal by strongly broadened absorption and luminescence lines of the Er^{3+} ion. The large absorption bandwidth provides the current stimulus for developing melilite hosts for diode-pumped rare earth ion lasers.

Theoretical Studies. To gain further insight of the bonding interactions in SLGO, theoretical calculations were made based on DFT methods. The calculated band structure of SLGO along high-symmetry points of the first Brillouin zone is plotted in Figure 4. It is clear that both the lowest conduction band and the highest valence band are localized at the G point; hence, SLGO is a direct band gap semiconductor. The calculated band gap of it is 3.265 eV. The bands can be assigned according to the total and partial DOS, as plotted in Figure 5. As seen in Figure 5, the peak localized around −30.0 eV is mostly contributed by Sr-5s and Ga-3d states. The peaks localized around −13.0 and −18.0 eV are mostly contributed by O-2s and Sr-4p states, while the peak localized around −1.0 eV is mainly contributed by the O-2p state. The peak from 4.9 to 8.0 eV in the conduction band is mainly contributed by La-6s, 5p and 5d states, mixing with a small amount of O-2p and Ga-4s states. In the vicinity of the Fermi level, namely, from −6.0 to 0 eV in the valence band, the O-2s and Sr-4p states are mainly involved, indicative of the strong covalent interactions of the Sr−O bond. The little overlapped electronic DOS of La and O, as well as Ga and O, demonstrates the existence of ion interactions of La−O and Ga−O bonds.

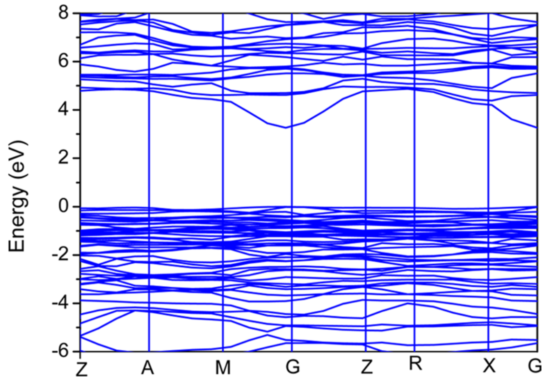

Figure 4. Calculated band structure of $SrLaGa_3O_7$.

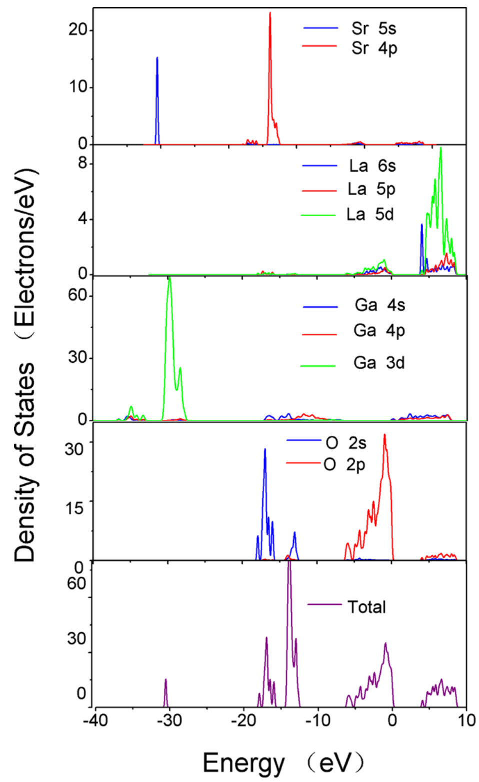

Figure 5. Electronic DOS of $SrLaGa_3O_7$.

Optical Properties. The room temperature absorption spectrum, near-infrared and mid-infrared waveband emission spectra as well as the up-conversion emission spectrum of the Er:SLGO crystal are shown in Figure 6. As seen in Figure 6a, the absorption spectrum consists of seven main absorption bands centered at 379, 488, 523, 652, 801, 978, and 1536 nm, which correspond to the transitions from $^4I_{15/2}$ to $^4G_{11/2}$ + $^2K_{15/2}$, $^4F_{7/2}$, $^2H_{11/2}$, $^4F_{9/2}$, $^4I_{9/2}$, $^4I_{11/2}$, and $^4I_{13/2}$, respectively. Among them, the wide absorption band centered at 978 nm with a full width at half-maximum (fwhm) of 24 nm matches the commercial 980 nm InGaAs laser diodes. According to the eq 1 in ref 1, the absorption cross section with a peak at 978 nm is calculated to be 1.01×10^{-21} cm^2. Figure 6b demonstrated the 1440−1630 nm waveband near-infrared emission spectrum of Er:SLGO crystals under 980 nm excitation. The emission around 1536 nm was observed in this crystal due to the transition of Er^{3+}:$^4I_{13/2} \rightarrow {}^4I_{15/2}$. The strong emission band with a peak at 2720 nm is observed in Figure 6c, corresponding to Er^{3+}:$^4I_{11/2} \rightarrow {}^4I_{13/2}$ in Er:SLGO crystal. This is the target emission, which would lead to the realization of an ∼2.7 μm laser of the Er:SLGO crystal. As seen in Figure 6d, the up-conversion emission spectrum within 500−700 nm of the Er:SLGO crystal is obtained under excitation of 980 nm. The observable green and red up-conversion emission bands which are centered at 548 and 669 nm are assigned to Er^{3+}: ($^2H_{11/2}$, $^4S_{3/2}$) → $^4I_{15/2}$ and $^4F_{9/2} \rightarrow {}^4I_{15/2}$ transitions, respectively. The green emission is much stronger than that of the red emission by nearly more than 15 times.

The fluorescence decay curves of Er^{3+}:$^4I_{13/2}$ and $^4I_{11/2}$ multiplets of the Er:SLGO crystal were measured at 1536 and 2720 nm excited by 980 nm, respectively, as shown in Figure 7. The decay curve from the $^4I_{13/2}$ level of the Er:SLGO crystal shows singly exponential decay behavior, whereas that of $^4I_{11/2}$ is multiexponential decaying owing to the existence of cross-relaxation. The fluorescence lifetimes were fitted and are presented in Figure 7 by using the method in our previous work.[3] It is noticed that the fluorescence lifetime of the $^4I_{11/2}$ state (0.71 ms) is shorter than that of the $^4I_{13/2}$ state (9.74 ms) in the Er:SLGO crystal; this might make the accumulated population in the $^4I_{11/2}$ level unable to relax quickly enough to maintain the necessary population inversion, and then would cause the ∼2.7 μm lasing transition to self-terminate automatically. However, the achievement of an ∼2.7 μm laser is due to the up-conversion process of Er^{3+} on the $^4I_{13/2}$ level: ($^4I_{13/2}$, $^4I_{13/2}$) → ($^4I_{15/2}$, $^4I_{9/2}$), which provides one channel for decreasing the populations of the lower $^4I_{13/2}$ level and increasing that of the upper $^4I_{11/2}$ level.

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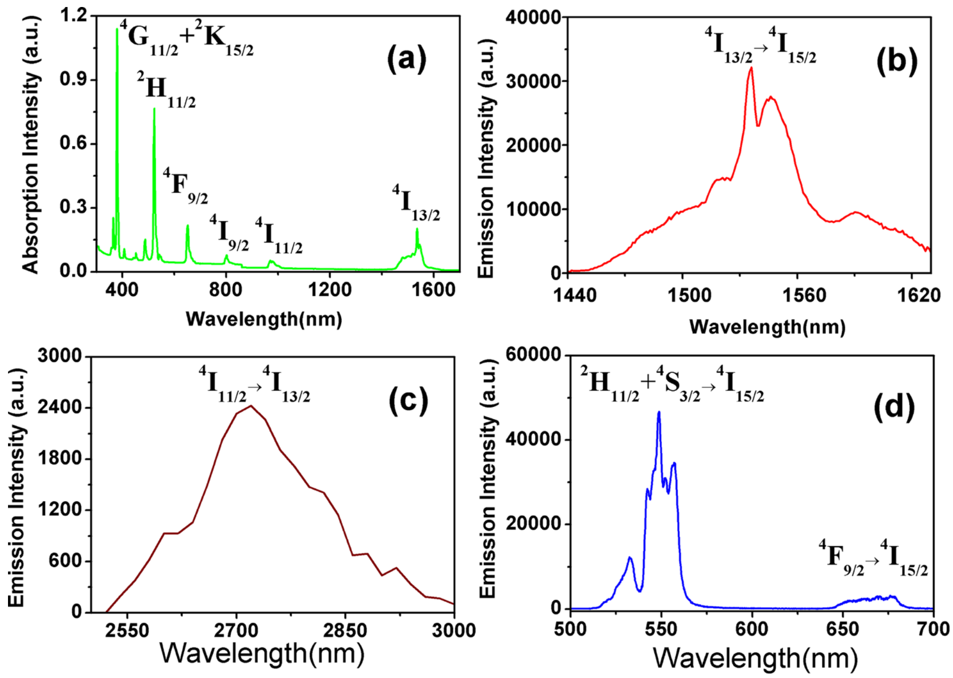

Figure 6. Absorption (a), near-infrared emission (b), mid-infrared emission (c), and up-conversion emission spectra (d) of Er^{3+}:$SrLaGa_3O_7$ crystal measured at room temperature.

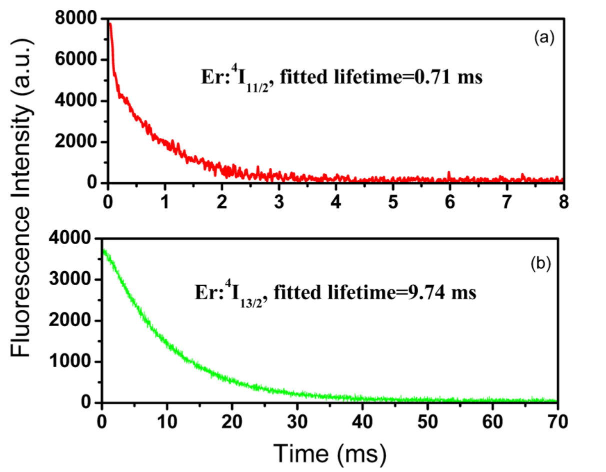

Figure 7. Fluorescence decay curves of Er^{3+}:$^4I_{13/2}$ and $^4I_{11/2}$ multiplets of Er:SLGO crystal measured at room temperature.

Besides, the rare earth ions with close energy as the Er:$^4I_{13/2}$ level can be codoped into the Er^{3+} activated crystal and thus increase the rate of population of ions on the lower level by resonant energy transfer, and thus inhibit the self-termination problem.[1−3,6−11,39,40]

The emission cross sections are calculated by the F-L equation[1]

$$\sigma_{em}(\lambda) = \frac{\beta\lambda^5}{8\pi cn^2\tau_r}\frac{I(\lambda)}{\int \lambda I(\lambda)\, d\lambda} \tag{1}$$

where $I(\lambda)/\int\lambda I(\lambda)\, d\lambda$ is the normalized line shape function of the experimental emission spectrum, β is the fluorescence branching ratio, c is the speed of light, n is the refractive index, and τ_r is the radiative lifetime. The maximum emission cross section of the Er:SLGO crystal is estimated to be 1.53×10^{-19} cm^2 at 2720 nm, while the maximum emission cross section of the Er:SLGO crystal is estimated to be 5.40×10^{-20} cm^2 at 1536 nm, which is much lower than that at 2720 nm. The obtained spectral parameters of it as well as the other two kinds of gallate crystals are listed in Table 1; it can be seen that it has comparable spectroscopic parameters to the Er:$SrGdGa_3O_7$ crystal as the other member of the ABC_3O_7 family, and better spectral characteristics than the commercial Er:YSGG crystal. The above-obtained spectroscopic parameters prove that the Er:SLGO crystal is a good candidate for achievement of ∼2.7 μm laser output.

■ CONCLUSION

In summary, the pure and Er^{3+}-doped SLGO single crystals with the melilite ABC_3O_7 structure were grown successfully by using the Czochralski technique. The structure of SLGO is demonstrated and analyzed. The theoretical calculations based on density functional theory (DFT) methods were performed on an SLGO crystal. The absorption and emission spectra of the Er:SLGO crystal were measured and analyzed. The absorption

Table 1. Spectroscopic Data for Er^{3+}:$SrGdGa_3O_7$ and Er^{3+}:$SrLaGa_3O_7$ Laser Crystals

crystals	σ_a (10^{-21} cm^2)	fwhm (nm)	σ_e (10^{-19} cm^2)	$\tau^4I_{11/2}$ (ms)	$\tau^4I_{13/2}$ (ms)	ref
Er:$SrGdGa_3O_7$	2.1@980 nm	31	1.64@2718 nm	0.62	10.81	40
Er:YSGG	1.56@790 nm	3	1.02@2640 nm	0.57	2.06	1
Er:$SrLaGa_3O_7$	1.01@978 nm	24	1.53@2720 nm	0.71	9.74	this work

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cross section is calculated to be 1.01 × 10^{-21} cm^2, and the emission cross sections with peaks at 1536 and 2720 nm are determined to be 5.40 × 10^{-20} and 1.53 × 10^{-19} cm^2, respectively. It is concluded that the Er:SLGO crystal with a disordered structure exhibits excellent optical properties and can act as a good candidate for a mid-infrared laser.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.cgd.6b00067.

Additional figures depicting experimental results (PDF)

Accession Codes

CCDC 1446141 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12, Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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